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Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 12	**Year: AY 2026-27**
TITLE : User-Defined Packages	

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Program:

StudentDetails.java

```
// File 1: student/StudentDetails.java
package student; // Package declaration
public class StudentDetails // Class must be public to be accessed
outside package
{
   public void showDetails(String name, int roll) // Public method
   {
       System.out.println("Student: " + name + " | Roll: " + roll);
   }
}
```

FacultyDetails.java

```
// File 2: faculty/FacultyDetails.java
package faculty; // Package declaration
public class FacultyDetails // Public class
{
   public void showDetails(String name, String dept) // Public method
   {
       System.out.println("Faculty: " + name + " | Department: " +
dept);
   }
```

```
}
```

CollegeManagementSystem.java

```
// File 3: CollegeManagementSystem.java (Main Program)
import student.StudentDetails; // Importing user-defined package class
import faculty.FacultyDetails; // Importing user-defined package class
import java.util.*;

public class CollegeManagementSystem
{
   public static void main(String args[]) // Main method
   {
       Scanner sc = new Scanner(System.in);
       System.out.print("Enter Student Name and Roll No: ");
       String sn = sc.next();
       int sr = sc.nextInt();
       System.out.print("Enter Faculty Name and Department: ");
       String fn = sc.next();
       String fd = sc.next();

       StudentDetails sObj = new StudentDetails(); // Creating instance
from student package
       FacultyDetails fObj = new FacultyDetails(); // Creating instance
from faculty package

       System.out.println("\n-- College Management System --");
       sObj.showDetails(sn, sr); // Calling package method
       fObj.showDetails(fn, fd); // Calling package method
       sc.close();
   }
}
```

CollegeManagementSystem.java

```
// File 1: library/Book.java
package library; // Package declaration
public class Book // Public class
{
   public void displayBook(int id, String title, String author, double
price) // Public method
   {
       System.out.println("Book ID: " + id);
       System.out.println("Title: " + title);
```

```
        System.out.println("Author: " + author);
        System.out.println("Price: $" + price);
    }
}
```

Library.java

```
// File 2: Library.java (Main Program)
import library.Book; // Importing user-defined package class
import java.util.*;
public class Library
{
    public static void main(String args[]) // Main method
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Book ID: ");
        int bid = sc.nextInt();
        sc.nextLine(); // Consuming newline
        System.out.print("Enter Book Title: ");
        String bt = sc.nextLine();
        System.out.print("Enter Book Author: ");
        String ba = sc.nextLine();
        System.out.print("Enter Book Price: ");
        double bp = sc.nextDouble();

        Book bObject = new Book(); // Creating instance from library
package

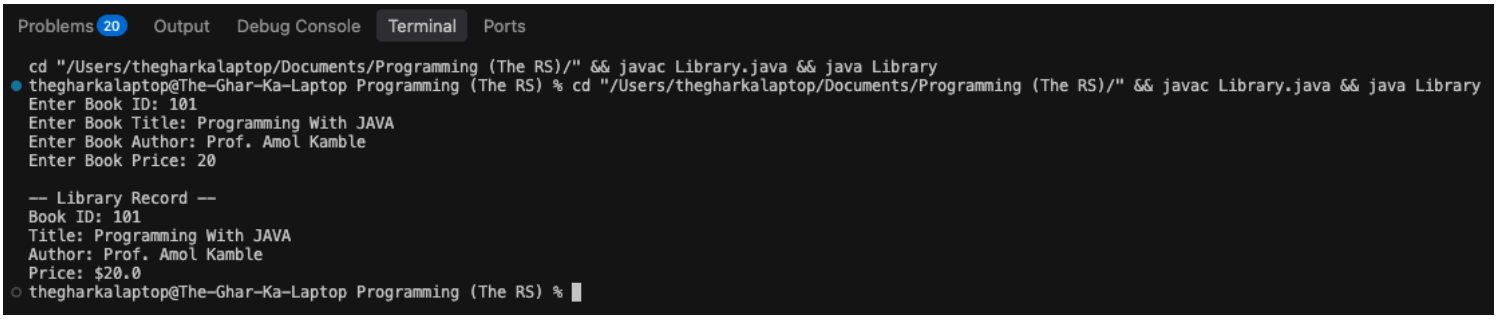
        System.out.println("\n-- Library Record --");
        bObject.displayBook(bid, bt, ba, bp); // Calling package method
        sc.close();
    }
}
```

Output:

CollegeManagementSystem.java



Library.java



Conclusion:

In this assignment, the concepts of code organization, namespace management, and access protection using user-defined packages were successfully implemented. By grouping related classes into separate directory structures like student, faculty, and library, the programs achieved modular application development, prevented naming conflicts, and established an organizational structure for larger Java projects.